

Design for Testability (DFT), DRC and LVS

Project Report

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Design for Testability (DFT), DRC and LVS

Abstract

This project focuses on understanding Design for Testability (DFT) concepts, scan chain insertion, stuck-at fault testing, Design Rule Check (DRC), and Layout Versus Schematic (LVS) verification. A 4-bit counter was designed and synthesized using Vivado. Although Vivado does not provide full ASIC DFT and LVS capabilities, it allows implementation, timing analysis, and DRC verification. The project demonstrates how DFT techniques improve fault detection and how DRC/LVS contribute to design correctness before fabrication.

Introduction

As digital circuits become more complex, testing and verification become critical to ensure correct functionality and manufacturability. Design for Testability (DFT) techniques improve fault detection during manufacturing testing. Scan chain insertion is one of the most widely used DFT methods. Additionally, physical verification techniques such as Design Rule Check (DRC) and Layout Versus Schematic (LVS) help ensure that the implemented design matches the intended design and satisfies fabrication constraints.

Objective

The objectives of this project are:

- To study Design for Testability (DFT) concepts.
- To understand scan chain insertion methodology.
- To analyze stuck-at fault models and test coverage.
- To perform synthesis and implementation using Vivado.
- To run Design Rule Checks (DRC).
- To understand the LVS verification process.

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Design Description

A 4-bit synchronous counter was selected as the sample design.

Verilog Code

```
module counter4(  
    input clk,  
    input rst,  
    output reg [3:0] q  
);  
  
always @(posedge clk or posedge rst)  
begin  
    if(rst)  
        q <= 4'b0000;  
    else  
        q <= q + 1'b1;  
    end  
  
endmodule
```

The counter increments its output value on every rising edge of the clock and resets to zero when the reset signal is asserted.

Design for Testability (DFT), DRC and LVS

Design for Testability (DFT)

Design for Testability is a methodology used to improve the controllability and observability of internal nodes within a digital circuit.

Advantages

- Improves fault detection.
- Reduces testing cost.
- Simplifies debugging.
- Increases product reliability.

Scan Chain Insertion

Scan chain insertion converts normal flip-flops into scan-enabled flip-flops.

Scan Signals

Signal	Description
Scan In (SI)	Serial input for test patterns
Scan Out (SO)	Serial output of captured data
Scan Enable (SE)	Enables scan mode
Clock	Controls shifting operation

Scan Chain Structure

Scan In → FF1 → FF2 → FF3 → FF4 → Scan Out

Modes of Operation

Functional Mode

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- Scan Enable = 0
- Circuit operates normally.

Scan Mode

- Scan Enable = 1
- Test data is shifted through flip-flops.

Since Vivado does not support ASIC scan insertion directly, scan chain implementation was studied conceptually.

Stuck-at Fault Analysis

A stuck-at fault assumes a signal is permanently fixed at logic 0 or logic 1.

Types

Stuck-at-0 (SA0)

Signal remains permanently at logic 0.

Stuck-at-1 (SA1)

Signal remains permanently at logic 1.

Example

If Q[0] of the counter is stuck at 0:

Expected Sequence:

0000 → 0001 → 0010 → 0011

Faulty Sequence:

0000 → 0010 → 0100 → 0110

The fault becomes observable during testing.

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Test Coverage

Test coverage measures the effectiveness of generated test patterns.

$$\text{Coverage (\%)} = (\text{Detected Faults} / \text{Total Faults}) \times 100$$

Example:

- Total Faults = 100
- Detected Faults = 98

$$\text{Test Coverage} = 98\%$$

Higher test coverage indicates better fault detection capability.

Vivado Design Flow

The following steps were performed using Vivado:

Step 1: Create Project

- Create RTL project.
- Select FPGA device.
- Add Verilog source file.

Step 2: Run Synthesis

- Check RTL correctness.
- Generate synthesized netlist.

Step 3: Run Implementation

- Placement and routing performed.
- Timing optimization completed.

Step 4: Generate Reports

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- Utilization Report
- Timing Summary
- DRC Report

Design Rule Check (DRC)

Purpose

DRC verifies that the design satisfies implementation and device-specific rules.

Common Checks

- Pin assignments
- Clock constraints
- Routing legality
- I/O standards
- Placement rules

Running DRC in Vivado

1. Open Implemented Design.
2. Select Reports → Report DRC.
3. Review generated report.

Sample Result

DRC SUMMARY

Total Violations : 0

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STATUS : PASS

The design passed all DRC checks successfully.

Layout Versus Schematic (LVS)

Purpose

LVS verifies that the physical layout matches the intended schematic.

LVS Flow

Schematic



Netlist Extraction



Layout



Extracted Layout Netlist



Comparison



PASS / FAIL

Typical LVS Checks

- Device matching
- Net matching
- Connectivity verification
- Pin correspondence

Design for Testability (DFT), DRC and LVS

Vivado does not provide ASIC LVS functionality. Therefore, LVS was studied conceptually.

Timing Analysis

Timing analysis was performed after implementation.

Timing Parameters

- Worst Negative Slack (WNS)
- Total Negative Slack (TNS)
- Setup Time
- Hold Time

Sample Timing Report

WNS = 18.091ns

TNS = 0.00 ns

Failing Endpoints = 0

The positive slack indicates that all timing requirements were satisfied.

Results

Parameter	Result
Synthesis	Successful
Implementation	Successful
DRC Status	PASS
Timing Status	PASS
Scan Chain	Conceptually Studied

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Stuck-at Fault Analysis Completed

LVS Verification

Studied Conceptually

Conclusion

The 4-bit counter design was successfully synthesized and implemented using Vivado. Design Rule Checks were performed and no violations were observed. Timing analysis confirmed that the design met all timing requirements. Scan chain insertion, stuck-at fault testing, and test coverage concepts were studied as part of Design for Testability. Although Vivado does not support full ASIC scan insertion and LVS verification, the project provided a strong understanding of DFT methodologies and physical verification concepts used in modern VLSI design flows.